

Cadillac WWTP MAHL (Benzidine, Beryllium, Phenol), submitted April 2026.

WRD Comments May 13, 2026. Claire Lowe.

- The study uses a plant removal efficiency (R_{POTW}) for phenol of 54.4% based on site specific removal data (page 5, Table 3). This removal efficiency is calculated based entirely on surrogate values as all influent and effluent results were non-detect, as stated in tables A13 and A14 (page 20), and the provided lab reports. The EPA Local Limits Development Guidance (EPA 833-R-04-002A) does not recommend the use of surrogate data if most data points are below the minimum level of quantitation of the method used. Please see EPA Local Limits Development Guidance Section 5.1.3, and Table 5-1 for more information. As stated in the comments on the previous version, it is recommended that the Appendix R literature value is applied instead of using surrogate values.
- The Removal efficiency (R_{prim}) applied in Table A3 for phenol is 18%. This doesn't agree with the stated removal efficiency used on p5, Table 3, which is the literature R_{prim} value of 8%.
- Table 11, page 8, states the calculated local limit for phenol as 2012 ug/l, but proposes to maintain the existing local limit as 1236 ug/L. While based on the calculations provided it appears that the existing phenol limit is protective of the POTW, the current limit appears more restrictive than is necessary, and may not be defensible as a technically based limit. It is recommended that a limit is proposed based on the MAHL calculations to ensure that if challenged the limit is defensible.
- The domestic flow (Q_{dom}) applied in table A5 is 0.8571 MGD. The domestic flow used in all other calculations agrees with the flow reported on page 10, Table A1 as 0.8362 MGD.
- The plant removal efficiency (R_{POTW}) for benzidine is reported as 87.7% as stated on page 5, Table 3, and used in subsequent calculations. Review of the removal calculations as given on page 20, Tables A13 and A14 show that the influent and effluent data points are averaged and rounded prior to calculating the removal. Rounding the data after calculation of the R_{POTW} produces lower removal efficiency of 87.5%. This does not result in a significantly different local limit but does alter the calculations.